Yuanfan Chen

646-963-8426 | yuanfan0504@gmail.com | github | linkedin | scholar

Education

Cornell Tech, Cornell University — New York, NY
Master of Engineering in Computer Science (GPA: 3.9) — *Aug. 2025 – May. 2026*

University of Toronto — Toronto, Canada
Honours Bachelor of Science, Computer Science Specialist (GPA: 3.9) — *Sep. 2020 – May. 2024*

Publications

- **Beyond Prediction: Tail-Aware Scheduling for LLM Inference**. Yueying Li, **Yuanfan Chen**, Jiayang Chen, Esha Choukse, Haoran Qiu, G. Edward Suh, Rodrigo Fonseca, Ziv Scully, Udit Gupta. **ICML 2026**. arXiv:2606.18431.
- **Making Sense of DPU Performance for Cloud Data Processing**. Jiasheng Hu, Chihan Cui, **Yuanfan Chen**, Philip A. Bernstein, Jialin Li, Qizhen Zhang. **SoCC 2026** (17th ACM Symposium on Cloud Computing). arXiv:2504.05536.

Open Source

vLLM | *ROCm, fp8 MoE*: Root-caused and fixed a production bug where uninitialized padded fp8 MoE expert weights made some replicas silently emit corrupted tokens (#55251).

LMCache | *Observability*: L1/L2 cache-hit attribution and Prometheus counters for the multi-process lookup path (#4734, #4962).

SGLang-Omni | *CUDA IPC, MPS*: Same-GPU weight sharing across data-parallel replicas (#1124); co-developed Restage placement ranking (#2134).

Experience

Machine Learning Infra Engineer — Aug 2026 – Present
*Character.AI: **Inference Optimization*** — *Bay Area, CA*

- Own production inference optimization for large-scale **multi-turn, long-context** chat at **10K+ QPS** on a **vLLM + LMCache** stack, serving models that differ in **architecture**, **attention mechanism**, and **input/output length distribution** across **thousands of GPUs**, including latest-generation **AMD Instinct** clusters; work spans **KV-cache offloading** and **kernel optimization**.
- Deployed a **hierarchical, distributed KV cache** (GPU → L1 → shared remote tiers) with **LMCache**, raising the **prefix-cache hit rate from 70% to 90%** and eliminating most long-context prefill recompute.
- Profiled AITER attention kernels with **AMD ROCm profilers** and traced Gemma-4 tail TTFT to head-512 attention layers at **∼6% MFU** (one workgroup exhausting a CU's 64 KB LDS, collapsing occupancy); retuned block/tile/warp configs for a **4.3× faster prefill kernel**, **32% faster cold prefill**, and **42–48% lower P99 TTFT** at equal load, doubling SLO-bound capacity per replica.
- Built a **real-traffic replay benchmark** to validate every optimization on production workload patterns before rollout.

Machine Learning System Researcher – S4AI Lab — Sep 2025 – Present
Cornell University — *New York, NY*

- Quantified that reasoning workloads cause **50% output-length misprediction** in LLM request schedulers; demonstrated via simulation that this misrouting degrades **P99 TTFT by up to 8×**, revealing a fundamental gap in production serving systems.
- Developed **PTA** (Prefill→Think→Answer) three-stage disaggregated serving; identified that each phase impacts UX differently, enabling **differential SLO constraints**; independently configured **DVFS frequency**, **KV-Cache allocation**, and **parallelism** per stage — preserving user-perceivable latency (**TTFAT**) via Answer-phase frequency protection under power caps while improving overall throughput through stage-level resource tuning.
- Built a quantitative **GPU frequency–performance model** by systematically profiling **DVFS** frequency vs. power, throughput, and latency curves on A100/H100; designed an **online optimization framework** that jointly schedules **LC/Flex/BE** request classes, maximizing throughput and QoS under **instantaneous power caps** and **cumulative energy budget** constraints.
- Applied **γ-Boost Queuing Theory** to design a prediction-free, tail-aware scheduler with **KV-cache-aware preemption**; reduced **tail TTFT/TTLT by 37–60%** while improving throughput by **1.01–1.12×** on Azure production traces.

Machine Learning Engineer — May 2025 – Jul 2025
*Tencent: **Distributed Training and Inference*** — *Beijing, CN*

- Designed **request-level admission control** with **batch-size stepwise profiling** for latency estimation and **SLA-aware priority scheduling**; maintained **100% goodput** under **2–10× QPS bursts** by enforcing per-class latency budgets across mixed-priority traffic.
- Identified **high prefix redundancy** in production traffic and adopted **SGLang with RadixAttention** for prefix caching; implemented **prefill-first chunking** over decode-first, reducing **P99 TTFT by ∼40%** without sacrificing decode throughput.
- Optimized CUDA attention kernels for **GQA** (A100) and **DeepSeek FlashMLA** sparse attention (H200); profiled with **Nsight Compute** to localize memory-hierarchy bottlenecks and redesigned thread-block tiling, shared-memory layout, and warp scheduling; achieved **∼20% throughput gain** on GQA and **∼15%** on FlashMLA over baselines.

Undergraduate Student Researcher – Far Data Lab — Jan 2024 – Feb 2025
University of Toronto — *Toronto, CA*

- Co-designed **dpBento**, a modular benchmark framework that standardizes performance evaluation across heterogeneous DPUs (**BlueField-2/3**, **OCTEON TX2**), from micro-level primitives to end-to-end cloud database offloading.
- Profiled **RocksDB** compaction and decompression paths end-to-end; offloaded compute-heavy stages to DPU, reducing **host CPU utilization by ∼40%** and achieving **∼3× decompression throughput** via hardware acceleration.

Academic Service

- **Invited Reviewer**: IEEE Journal on Selected Areas in Information Theory (JSAIT) – Special Issue on Energy and Data Efficiency in AI (2026). Reviewed submissions focusing on cutting-edge LLM inference systems.